SMART INDIA HACKATHON 2026 · PS 26070

AI/ML System for Identification, Classification & Prediction of Tropical Cyclones using Multi-Source Satellite Data

Prototype guide — what it does, results, demo & video script, PPT content, and how to take it live

Problem Statement ID	26070
Organisation	Ministry of Earth Sciences (MoES) — India Meteorological Department (IMD)
Theme / Category	Disaster Management · Software
Team	NerveAi (Team ID: _____)
Prototype	Web dashboard (Next.js) + FastAPI backend + 7 trained AI models · runs locally, Docker-ready

Contents

1. What the prototype does and the problem it solves
2. How it solves the problem (end-to-end flow)
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7. Demo walkthrough (click path with deep links)
8. 5-minute video: timeline, screens and voice-over
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11. Going live: hosting the backend and frontend
12. Judge Q&A preparation, limitations and future scope

All diagrams are also saved as separate 1600×900 (16:9) PNG files in `docs/diagrams/` and screenshots in `docs/screenshots/` for direct use in PowerPoint.

1 · What the prototype does

It turns raw satellite imagery of the North Indian Ocean into early, explained and actionable cyclone guidance: it **finds** cyclonic systems in the whole-basin image, **classifies** their intensity on IMD's scale, **warns** of genesis and rapid intensification, **predicts** track and intensity with a calibrated uncertainty cone, and **converts** the forecast into landfall, exposure and district-level alerts in the Common Alerting Protocol (CAP) format.

The problem it addresses

- **Open ocean has almost no ground observations** — satellites are the main source of truth for storms over the Bay of Bengal and Arabian Sea.
- **Rapid intensification (RI)** — winds rising ≥30 kt in 24 h — is the hardest forecast problem and leaves the least time for evacuation (Ockhi 2017, Biparjoy 2023). RI is rare (≈1 in 15 observations), so naive models ignore it.
- **Data scarcity:** the North Indian Ocean has far fewer historical storms than the Pacific, which limits machine learning.
- **Uncertainty is rarely communicated:** a single forecast line looks precise but misleads decision-makers.
- **The last mile:** a forecast only saves lives when it reaches the right district and fishing community in their language.

Capabilities at a glance

Capability	How	Evidence (held-out data)
Identification (No cyclone / Cyclone, location)	ResNet-18 heatmap detector scans the entire 0–30°N, 40–100°E IR + water-vapour scene; detections linked into tracks	6 of 6 demo storms found 24–42 h before Cyclonic Storm stage; 79–94% of fixes detected
Classification (IMD D → SuCS)	EfficientNet-B0 on IR brightness temperature with wind, Dvorak T-number and class heads; Grad-CAM explanation	Wind error 6.2–7.4 kt MAE on unseen 2016 NIO storms
Genesis outlook	LightGBM on track history + ERA5 environment	ROC-AUC 0.89 (2020–25)
Rapid-intensification warning	Class-weighted, calibrated LightGBM with SHAP reasons	8 of 8 RI storms of 2020–25 warned; ROC-AUC 0.85
Track & intensity prediction 6–72 h	LSTM + Monte-Carlo dropout, cone calibrated to 67%	Cone coverage 66%; 12–18% better than persistence
Impact & alerts	50-member ensemble landfall, WorldPop exposure, district staging, sea areas, CAP 1.2 XML, SMS (EN/HI/GU)	Biparjoy replay: 12 Gujarat districts staged on 13 June 2023

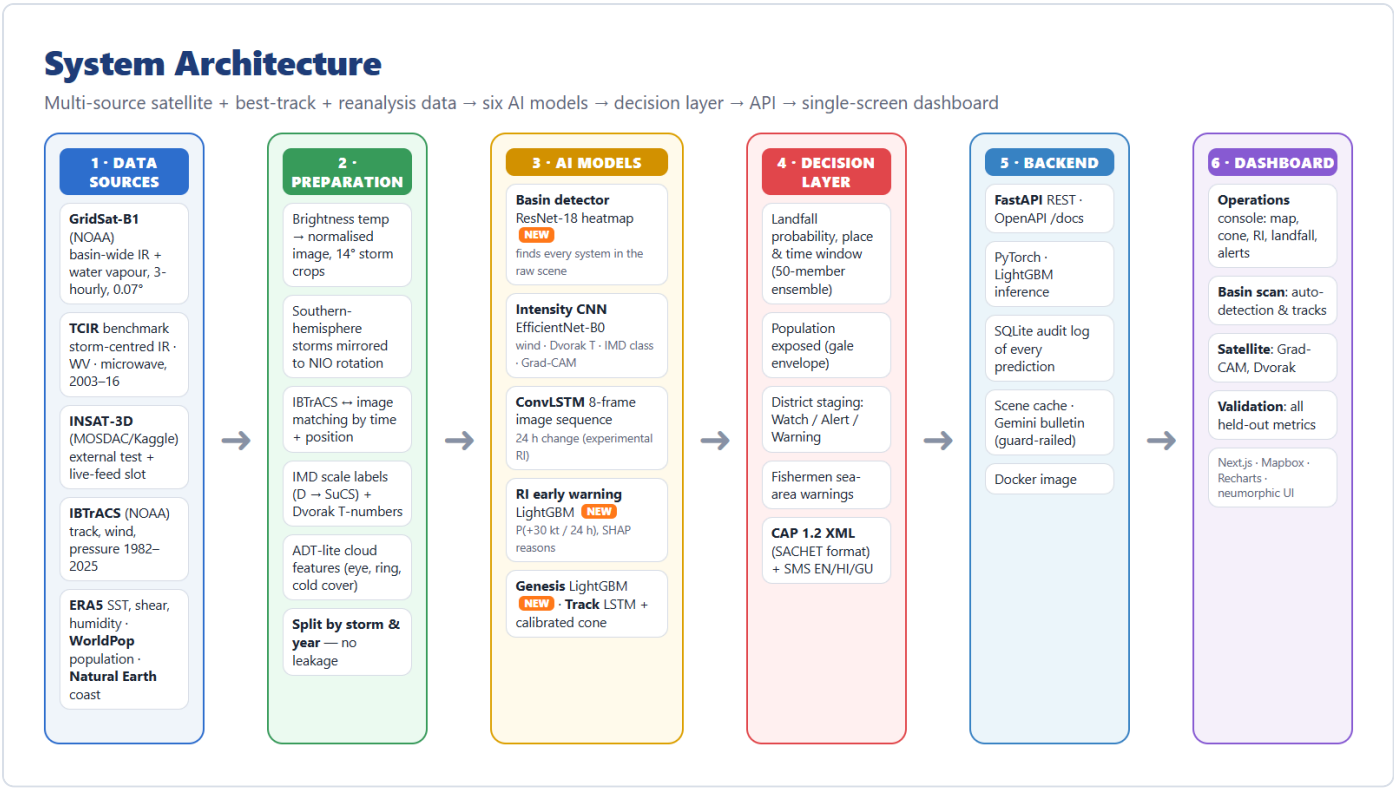


Figure 1 — System architecture (docs/diagrams/01_system_architecture.png)

2 · How it solves the problem



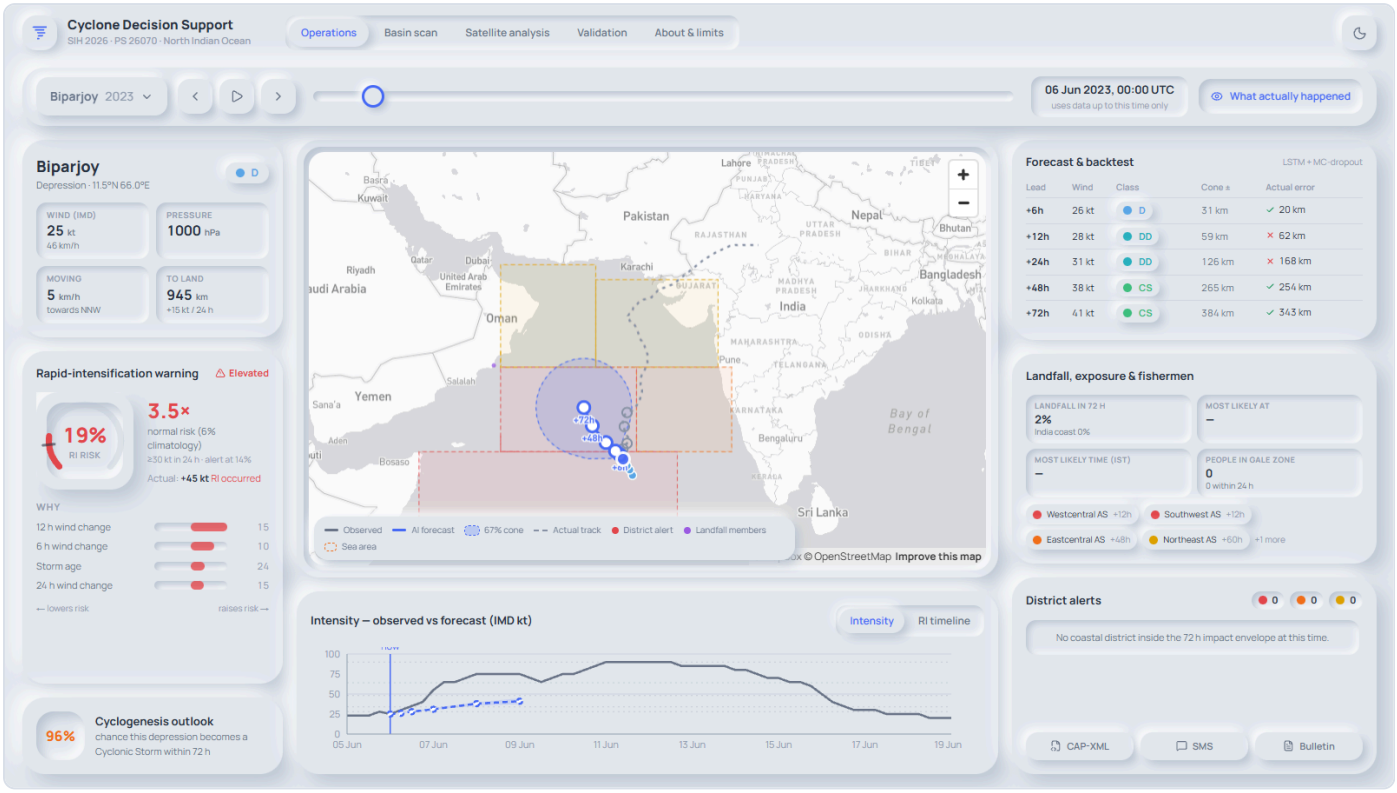
Figure 2 — From one satellite image to one actionable alert

1. **Satellite scene.** Every 3 hours the whole-basin infrared (IR) and water-vapour (WV) image is read. Cold cloud tops = deep convection.
2. **Identify.** The basin detector marks every cyclone or depression centre and links them over time into tracks. A track is "confirmed" after persisting 15 h, which filters most false alarms.
3. **Classify.** A 14°×14° crop around each centre goes to the intensity CNN, which returns three cross-checked estimates — wind speed, Dvorak T-number (converted with the standard Dvorak table) and IMD category — plus a Grad-CAM heat map showing which clouds drove the estimate.
4. **Warn early.** A genesis model estimates whether a young depression will become a Cyclonic Storm within 72 h; the RI model estimates the probability of a ≥30 kt jump in the next 24 h and lists the top reasons (SHAP).
5. **Predict.** The track model forecasts position and intensity at +6/12/24/48/72 h. Monte-Carlo dropout gives a spread which is scaled so that 67% of real positions fall inside the cone — the same convention as operational cones.
6. **Assess impact.** Fifty calibrated forecast members give a landfall probability, most likely country/district and time window; population inside the damaging-wind envelope is counted from WorldPop; fishermen warnings are issued per sea area.
7. **Alert.** Coastal districts are staged Watch / Alert / Warning and coloured Yellow / Orange / Red; a CAP 1.2 XML alert (the format NDMA's SACHET system ingests) and short SMS texts in English, Hindi and Gujarati are generated.

Key design principle — replayable, no look-ahead. Every screen uses only data available at the selected time, so any storm since 1990 can be replayed and compared with "what actually happened". This is how the prototype proves it works rather than just showing a dashboard.

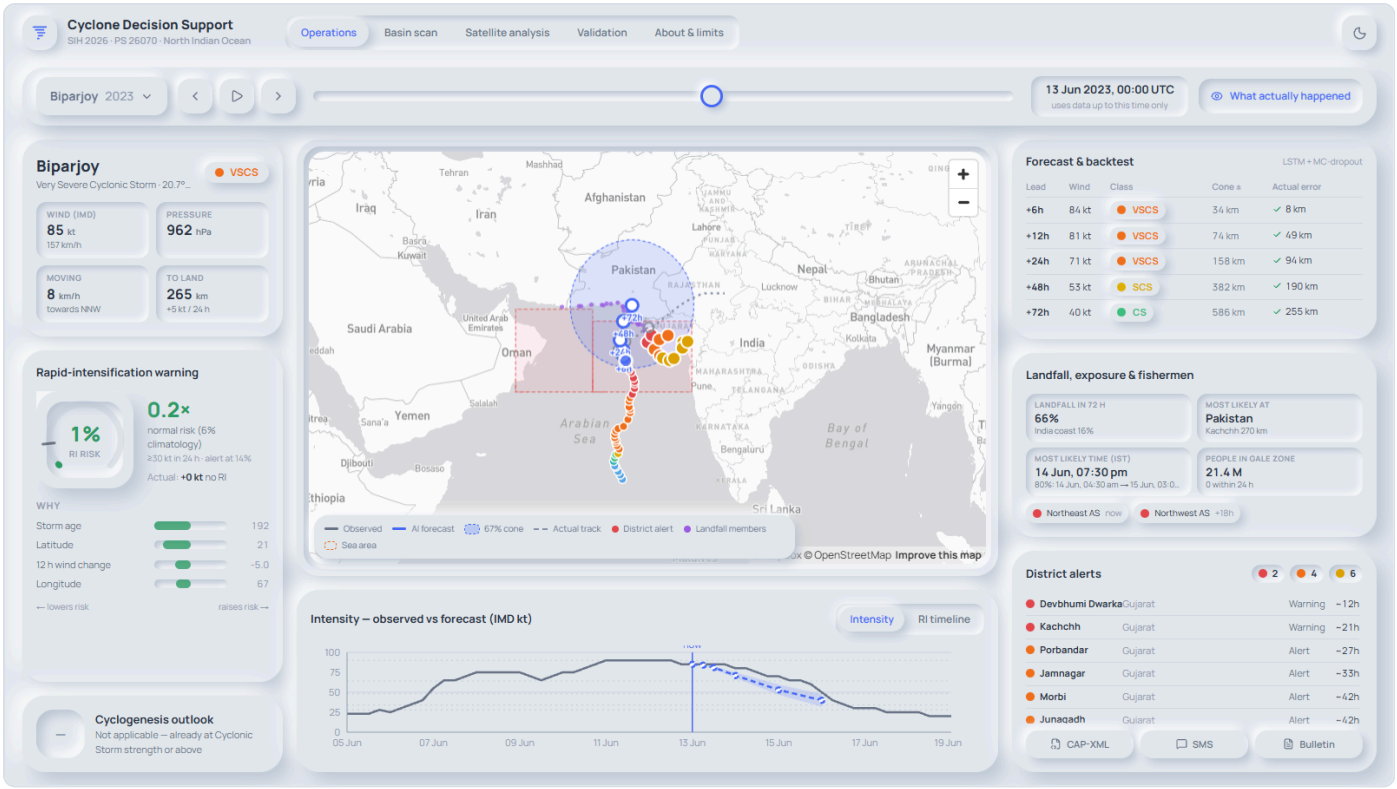
3 · How the prototype shows it — the screens

Operations console (single screen)



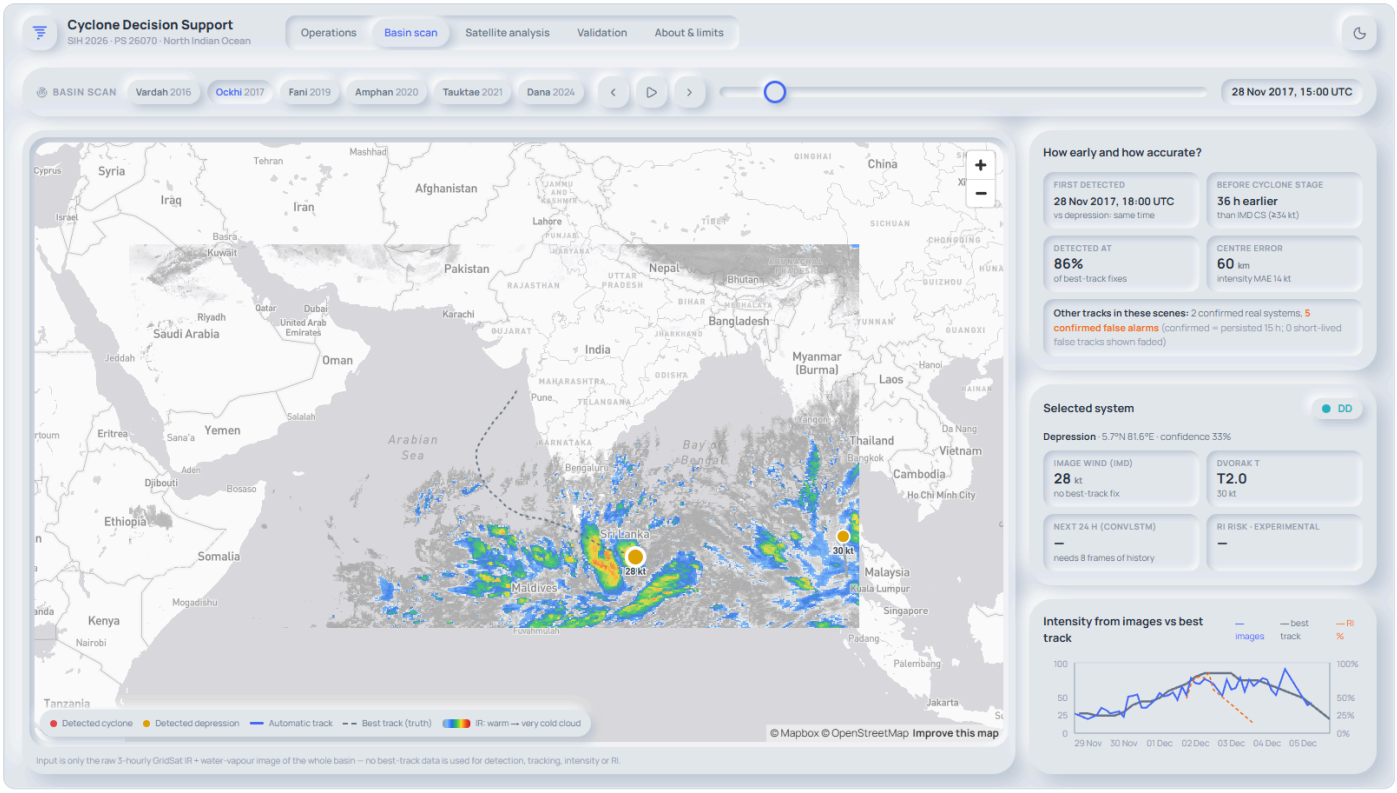
Biparjoy, 6 June 2023 00 UTC: still a 25 kt Depression, RI warning at 3.5× normal risk — it then gained 45 kt in 24 h.

Left: current state, RI gauge with reasons, genesis outlook. **Centre:** map with observed track, AI forecast, 67% cone, actual future track (toggle), landfall members, sea areas, district alerts; chart strip (intensity or RI timeline). **Right:** forecast & backtest table, landfall/exposure/fishermen, district alerts with CAP-XML, SMS and Bulletin buttons.



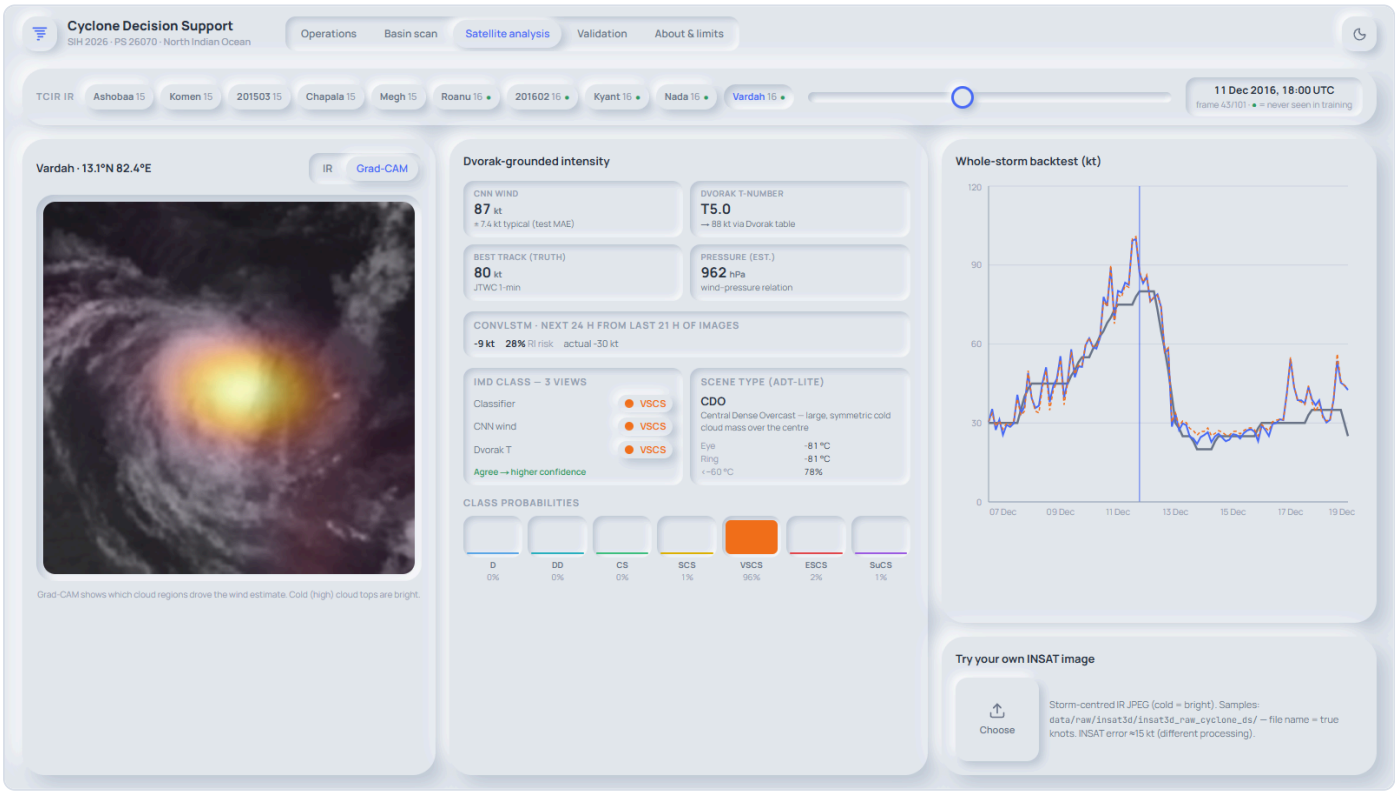
Biparjoy, 13 June 2023: 12 Gujarat districts staged, 66% landfall probability, 21.4 M people in the gale zone.

Basin scan — identification from raw imagery



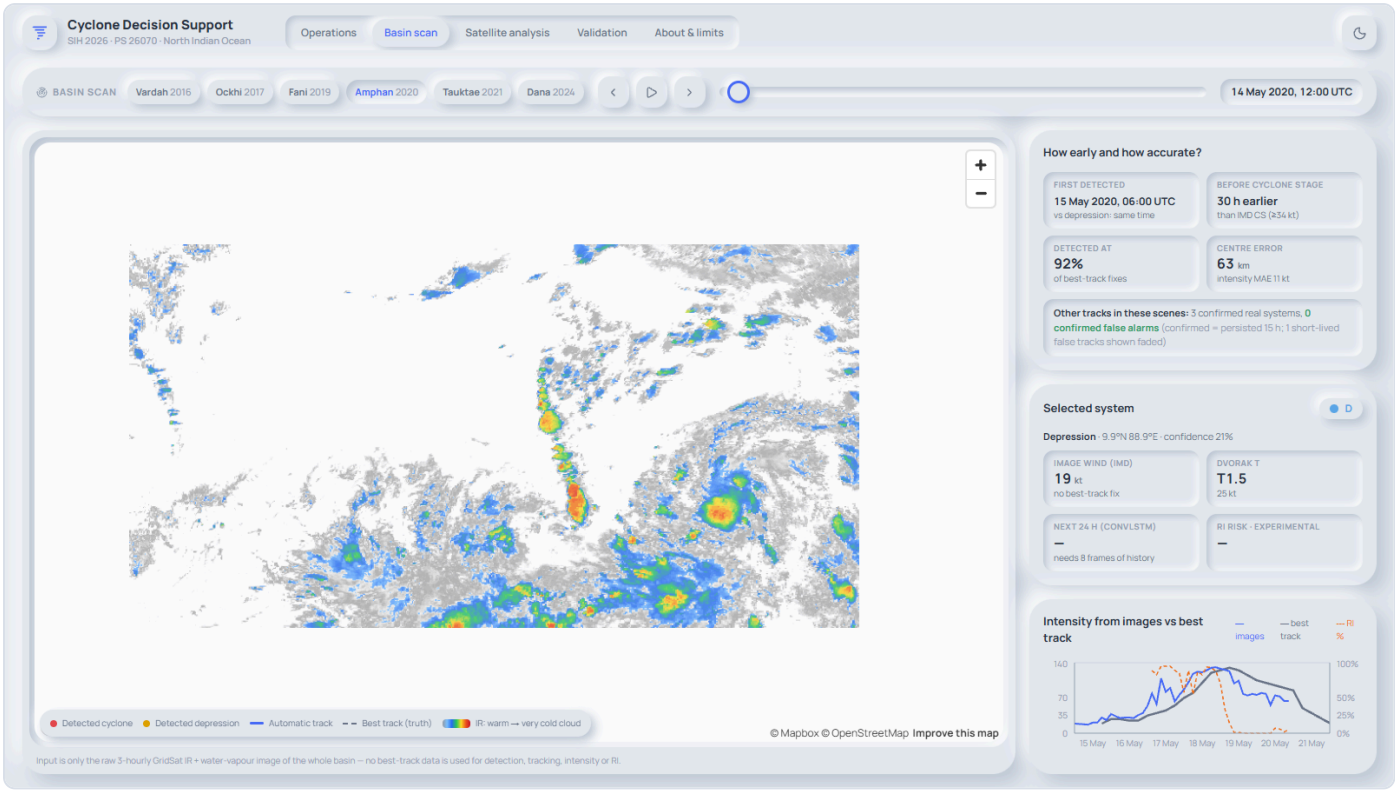
Ockhi 2017: the enhanced IR image of the whole basin; red/yellow dots are detections, blue lines automatic tracks, dashed line the best track (truth).

Satellite analysis — explained intensity



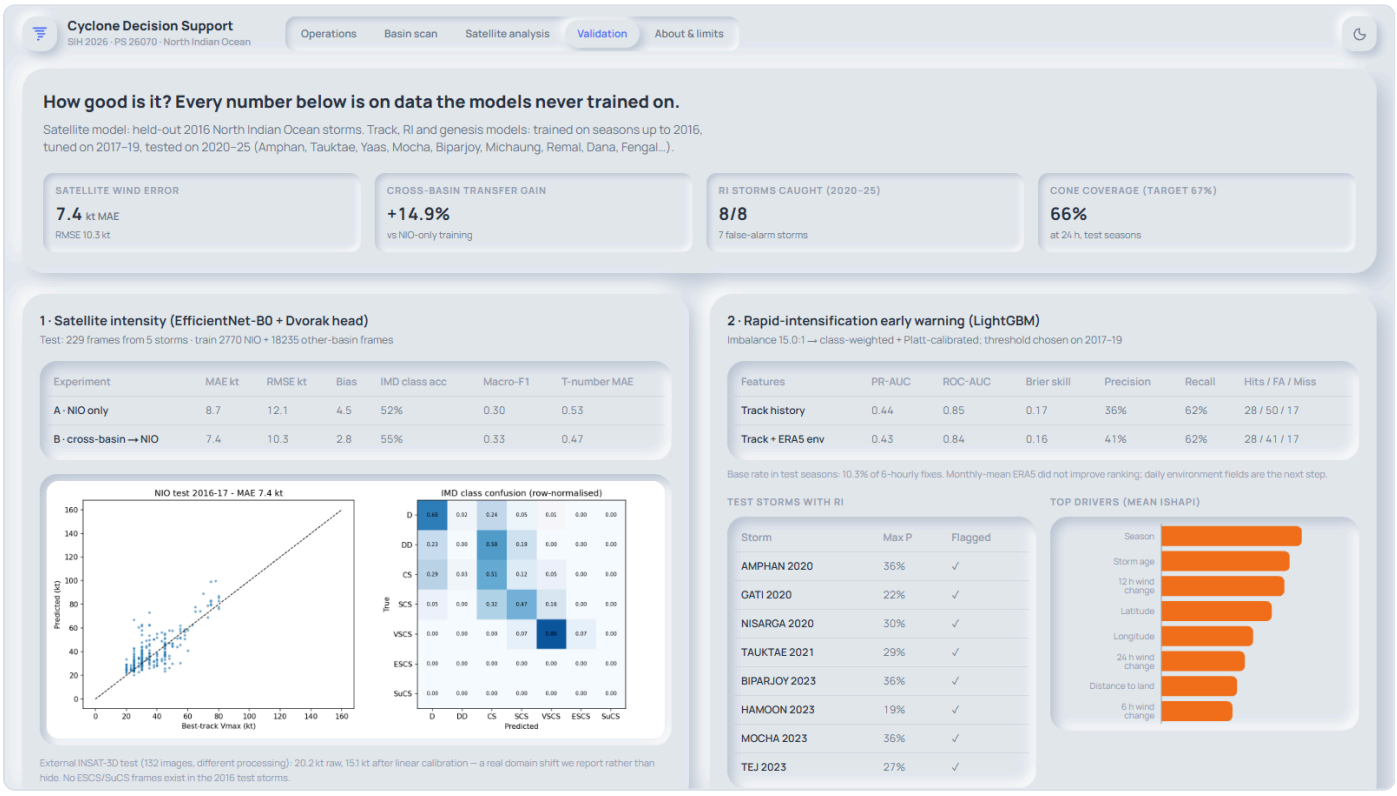
Vardah 2016 (never seen in training): Grad-CAM, CNN wind vs Dvorak T-number vs IMD class, scene type, whole-storm backtest.

Basin scan — a Super Cyclonic Storm



Amphan 2020 in the Bay of Bengal: detected in 92% of fixes, 30 h before Cyclonic Storm stage.

Validation — every number on held-out data



The Validation page lists all metrics, including the negative results (see section 5).

4 · How it matches the problem statement's expectations

The PS asks for an AI/ML system for **identification, classification and prediction** of tropical cyclone patterns using **multi-source satellite data**. The table maps every item of the team's implementation plan to what was built.

PS / plan expectation	What the prototype does	Where to see it	Status
Identification of cyclones from satellite images	Whole-basin detector finds every system and tracks it; 24–42 h before Cyclonic Storm stage on 6 demo storms	Basin scan	Done
Label images No Cyclone / Cyclone	Detector trained on storm scenes plus storm-free scenes (negatives); classes cyclone (≥34 kt) and depression (17–33 kt)	Basin scan, Validation §5	Done
Classify cyclone intensity (IMD categories)	CNN wind + Dvorak T-number + IMD class (D, DD, CS, SCS, VSCS, ESCS, SuCS), three views cross-checked	Satellite, Basin scan	Done
Multi-source satellite data	IR + water vapour (GridSat-B1), IR + WV + passive microwave (TCIR), INSAT-3D external test; plus IBTrACS, ERA5, WorldPop	Validation §6	Done
Match satellite images with IBTrACS by time + location	Every TCIR/GridSat frame matched to the IBTrACS storm (e.g. TCIR 201605I → Vardah)	Satellite	Done
Focus on Arabian Sea and Bay of Bengal	Scene box 0–30°N, 40–100°E; NIO-specific fine-tuning; Arabian-Sea flag in models	All pages	Done
CNN / EfficientNet for detection & classification	EfficientNet-B0 intensity model; ResNet-18 basin detector	Satellite, Basin scan	Done
LSTM / ConvLSTM to predict movement and intensity (6–24 h)	LSTM track + intensity to 72 h with calibrated cone; ConvLSTM on 8 satellite frames for 24 h change	Operations	Done
Input latest image → location, category, wind speed, predicted path	Basin scan gives location, category and wind from the image alone; the path forecast uses the track history of that system	Basin scan + Operations	Partly
INSAT-3D/3DR/3DS from MOSDAC	INSAT-3D images (MOSDAC-sourced, Kaggle) used as an external test; live MOSDAC ingest is designed but pending account approval. Training used GridSat/TCIR geostationary IR (same physical channel)	Satellite upload, Validation §1	Partly
Dashboard: React + FastAPI + map	Next.js (React 19) + FastAPI + Mapbox; single-screen console	All pages	Done
Display satellite image, cyclone info, predicted trajectory	Basin IR overlay, Satellite page, cyclone state card, forecast track + cone	All pages	Done

Beyond the PS: genesis outlook, rapid-intensification warning with reasons, calibrated uncertainty, landfall window, population exposure, fishermen sea-area warnings, CAP 1.2 alerts and multilingual SMS — the pieces an RSMC forecaster, SDMA officer or district collector actually acts on.

5 · Results and success rates

Cyclone models are not judged by a single "accuracy %". Each is measured the way forecasters measure it, on storms or seasons the model never saw during training.

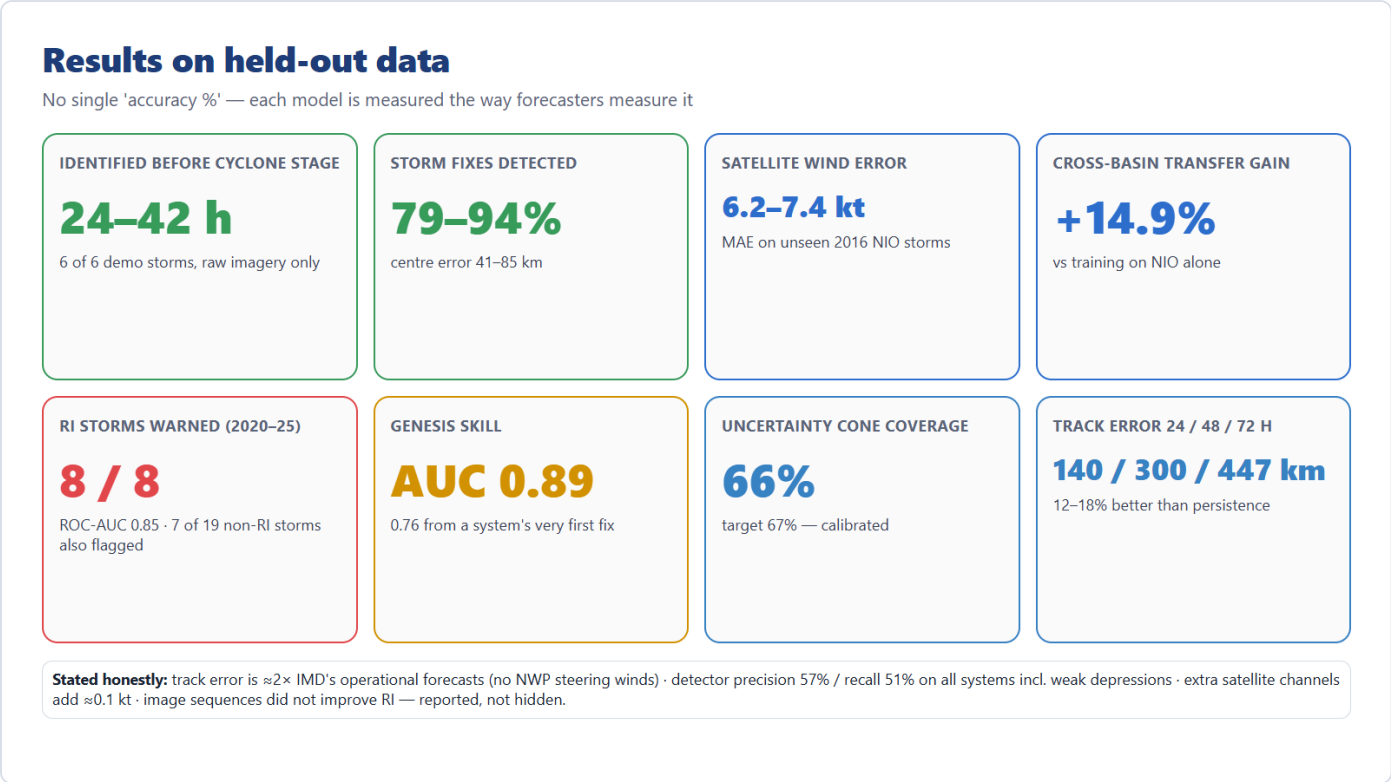


Figure 3 — Results at a glance

Model	Test set	Result	In plain words
Intensity CNN (IR)	5 NIO storms of 2016, 229 frames	MAE 7.4 kt, RMSE 10.3 kt; IMD class accuracy 55%; T-number error 0.47	Wind within ~7 kt on average; class often off by one neighbouring category
Cross-basin transfer	same	MAE 8.7 → 7.4 kt (**+14.9%**)	Learning from other oceans helps India's data-scarce basin
Multi-channel CNN	same	MAE 6.2–6.5 kt; extra channels change error by ≈0.1 kt	Better training; IR carries almost all the signal
INSAT-3D external test	132 labelled INSAT-3D images	20.2 kt raw → 15.1 kt after calibration	Different image processing = domain shift; retrain on native MOSDAC data
Basin detector	189 scenes 2017–24	Precision 57%, recall 51% (all systems ≥17 kt), centre error 69 km	Finds most systems; weak depressions are hardest
Image-only pipeline	6 demo storms	Detected 24–42 h before CS stage; 79–94% of fixes; 41–85 km; 14 real vs 6 false confirmed tracks	Spots storms 1–2 days before they are cyclones
RI early warning	2020–25, 436 fixes, 45 RI	ROC-AUC 0.85, PR-AUC 0.44 (base rate 0.10), recall 62%, precision 36%; 8/8 RI storms flagged, 7/19 non-RI storms also flagged	Catches every RI storm, at a cost of some false alarms

Genesis	2020–25	ROC-AUC 0.89; 0.76 from a system's first fix	Good at telling which depressions will become cyclones
Track + cone	2020–25, 571 forecasts	140 / 300 / 447 km at 24 / 48 / 72 h; 12–18% better than persistence; cone coverage 66% (target 67%)	Honest cone; track ≈2× IMD operational error
ConvLSTM sequence	All-basin validation	24 h change MAE 13.3 vs 13.9 kt without images; RI PR-AUC 0.30 vs 0.42 without images	Images did not help RI → shown as experimental

How to talk about "success rate" in front of judges. Lead with the three strongest, verifiable numbers: **identified 24–42 h before cyclone stage** (6/6 storms), **8 of 8 rapid-intensification storms warned** (2020–25), and a **calibrated uncertainty cone (66% vs 67% target)**. Then state the limits yourself before a judge asks: track error ≈2× IMD's, detector recall 51% for weak depressions, INSAT domain shift. Never quote a single "accuracy %".

6 · What we built differently

What we built differently

Compared with the common PS 26070 pattern (CNN detects → LSTM predicts → map)

	Typical build	Our prototype
Identification	✗ Runs on a pre-cropped image of a known storm	✓ Scans the whole basin, finds & tracks systems automatically
Early warning	✗ Class label after the storm exists	✓ Genesis probability + rapid-intensification warning with reasons
Intensity	✗ Bare CNN class	✓ Dvorak T-number + IMD class, 3 cross-checked views, Grad-CAM
Track	✗ Single deterministic line	✓ Calibrated 67% uncertainty cone (checked: 66%)
Data scarcity	✗ Train on NIO only	✓ Cross-basin pretraining: +14.9%
Impact	✗ Map only	✓ Landfall window, people exposed, fishermen sea areas
Alerting	✗ Generic SMS mock	✓ CAP 1.2 XML (SACHET format) + EN/HI/GU SMS with segment counts
Proof	✗ Live demo on chosen inputs	✓ Replay any storm since 1990 against what actually happened
Honesty	✗ Accuracy % on random split	✓ Storm/year splits, false alarms & limits shown on screen

Figure 4 — Compared with the common PS 26070 pattern

- **Starts before the storm exists:** basin-wide detection and a genesis probability, not a classifier that needs a pre-cropped storm image.
- **Targets the hardest sub-problem:** a dedicated, imbalance-aware rapid-intensification warning with human-readable reasons.
- **Operationally grounded intensity:** Dvorak T-number and IMD category alongside the CNN wind, with Grad-CAM.
- **Honest uncertainty:** the cone is calibrated and its coverage is measured.
- **Data scarcity addressed:** cross-basin transfer learning, measured +14.9%.
- **Closes the loop:** landfall, exposure, fishermen warnings, CAP 1.2 XML and multilingual SMS.
- **Proof, not just a demo:** replay any storm since 1990 against what actually happened; negative results published.

Careful wording: say "not seen in the public PS 26070 prototypes we reviewed" rather than "no one else does this".

Who uses it — and the decision each screen supports

Every panel on the dashboard is labelled with its intended user

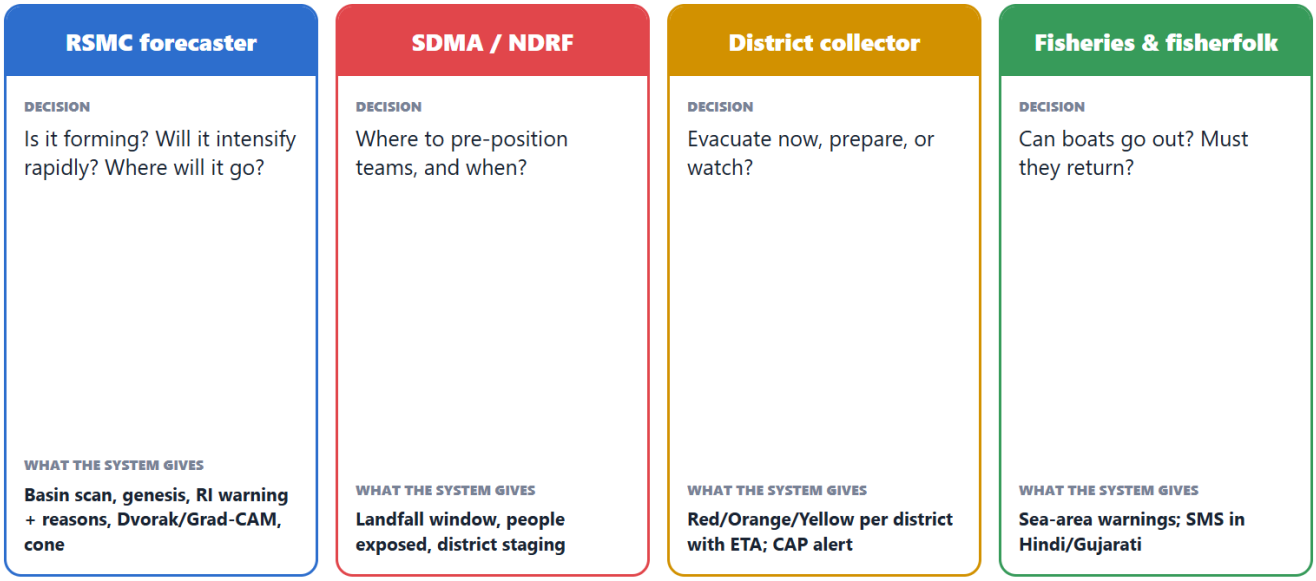


Figure 5 — Users and the decision each screen supports

7 · Demo walkthrough (click path)

Start the app with `.\run_dev.ps1` (or open the deployed URL). Deep links jump straight to each moment — replace `localhost:3000` with the deployed address when live.

#	Open	What to do	What to point out
1	`localhost:3000/basin?storm=OCKHI`	Press ▶ play; click a red/yellow detection	Detected from raw imagery; "How early" card: 36 h before CS stage; false alarms shown
2	`localhost:3000/basin?storm=AMPHAN`	Drag the slider to the peak	Image-only intensity (blue) follows best track (grey); Super Cyclonic Storm
3	`localhost:3000/satellite`	Toggle IR / Grad-CAM; move the frame slider	Three intensity views agree; Dvorak T-number; ConvLSTM outlook
4	`localhost:3000/?storm=BIPARJOY&time=2023-06-06T00:00`	Point to RI gauge, then "Actual: +45 kt"	RI warned while still a 25 kt Depression; SHAP reasons; genesis 96%
5	same page	Toggle "What actually happened"; switch chart to RI timeline	Cone vs real track; red bars = RI actually happened
6	`localhost:3000/?storm=BIPARJOY&time=2023-06-13T00:00`	Open CAP-XML, then SMS, then Bulletin	12 districts; landfall 66%; 21 M exposed; CAP 1.2; Hindi/Gujarati SMS segments
7	`localhost:3000/validation`	Scroll sections 1–8	All held-out; negative results included
8	`localhost:8000/docs`	Try `/api/storms` live	Real API, not hard-coded; every prediction logged

Other good storms to replay: Mocha 2023, Amphan 2020, Tauktae 2021 (search box in the storm menu), and on Basin scan: Fani 2019, Tauktae 2021, Dana 2024, Vardah 2016.

Before any live demo: open every page once to warm the caches (first basin scan of a storm takes 20–40 s on CPU; afterwards it is instant). Keep a recorded video as backup.

8 · 5-minute video: timeline, screens and voice-over

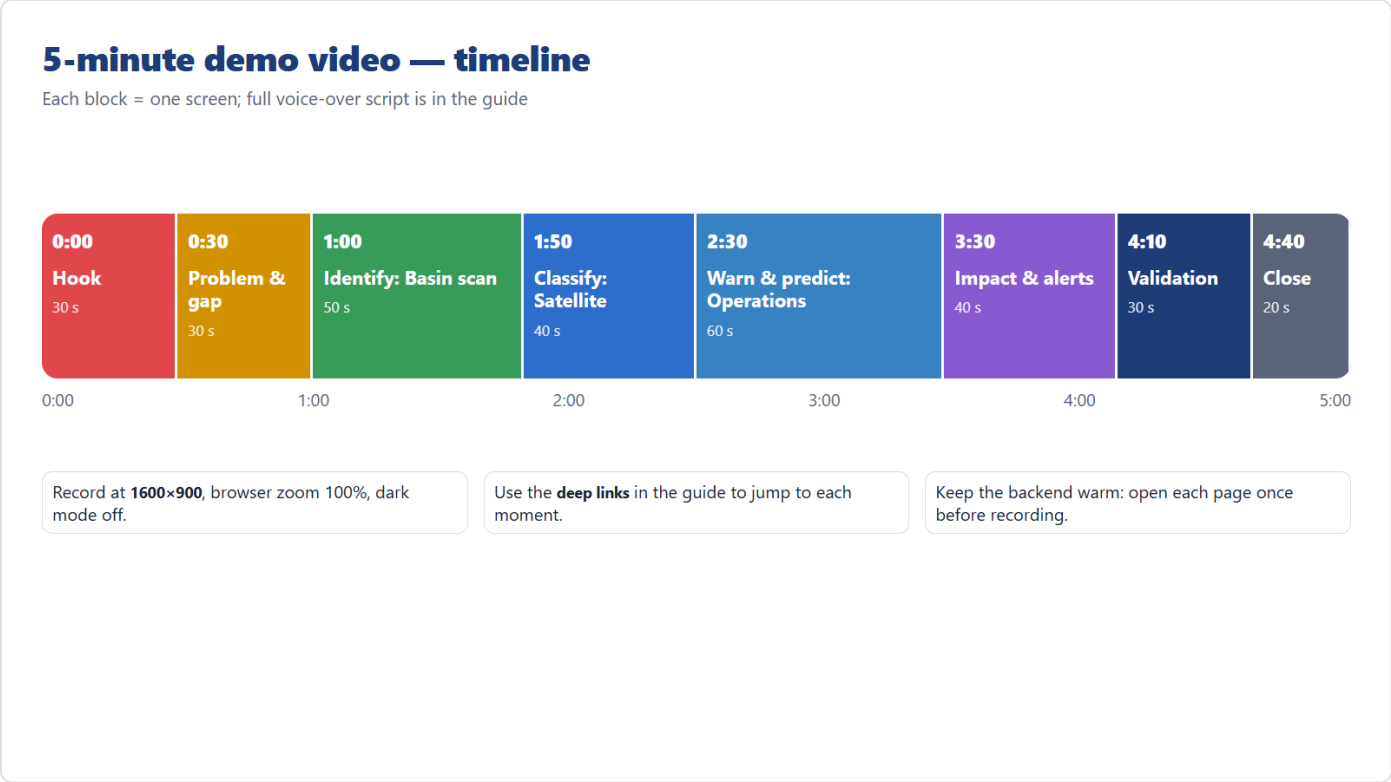


Figure 6 — Video timeline (≈650 words of narration at a calm 130 words/minute)

Time	Screen / link	On-screen action	Voice-over
0:00–0:30 Hook	Operations — `?storm=BIPARJOY&time=2023-06-06T00:00`	Slow zoom on the 25 kt Depression, then the "Actual +45 kt" line	*"In June 2023, Cyclone Biparjoy sat in the Arabian Sea as a weak depression. Within 24 hours its winds jumped by forty-five knots. Rapid intensification like this is the hardest problem in cyclone forecasting — and it leaves coastal districts the least time to act. We are Team NerveAi, and this is our AI system for Problem Statement 26070 from the Ministry of Earth Sciences and IMD."*
0:30–1:00 Problem & gap	Diagram 02 (end-to-end flow) or About page	Reveal the 7 steps one by one	*"Most AI cyclone projects classify a storm once it already exists and draw one track line. Forecasters need more: to spot systems early, to know when a storm will suddenly strengthen, to see how uncertain the forecast is — and to deliver it in the format disaster managers already use. Our prototype does all of this from multi-source satellite data."*

1:00–1:50 Identify	Basin scan — `/basin?storm=OCKHI`	Press ▶; pause when the first detection appears; point to the "How early" card	*"This is the raw infrared and water-vapour image of the entire North Indian Ocean, every three hours. No track data goes in. Our detector finds every system, links detections into tracks and estimates intensity from the image alone. For Cyclone Ockhi in 2017, the system identified the depression thirty-six hours before it became a cyclonic storm. Across six demo storms, identification came twenty-four to forty-two hours before cyclone stage — and we show the false alarms too."*
1:50–2:30 Classify	Satellite — `/satellite` (Vardah)	Toggle IR → Grad-CAM; hover the three class views	*"For intensity, a CNN trained with cross-basin transfer learning reads the infrared image and gives three independent estimates: a wind speed, a Dvorak T-number mapped through the standard Dvorak table, and an IMD category. When they agree, confidence is higher. Grad-CAM shows which clouds drove the estimate. On storms it never saw, the average error is about seven knots."*
2:30–3:30 Warn & predict	Operations — back to 6 June; switch chart to "RI timeline"; press ▶ for a few steps	Point to RI gauge and reasons, the cone, the dashed actual track and the genesis card	*"Back to Biparjoy on the sixth of June. The system gives a nineteen percent chance of rapid intensification — three and a half times the normal risk — and explains why: the strengthening in the last twelve and twenty-four hours. What actually happened: plus forty-five knots. On the 2020 to 2025 test seasons it warned for all eight rapidly intensifying storms. The track forecast comes with a calibrated uncertainty cone — real positions fell inside it sixty-six percent of the time, against a sixty-seven percent target — and a genesis model estimates whether a depression will become a cyclone at all."*
3:30–4:10 Impact & alerts	Operations — `?storm=BIPARJOY&time=2023-06-13T00:00`	Point to landfall card and district list; click CAP-XML, then SMS	*"A week later, the forecast becomes decisions: a landfall probability and time window, twenty-one million people inside the damaging-wind zone, fishermen warnings for each sea area, and red, orange and yellow alerts for twelve Gujarat districts. One click produces a Common Alerting Protocol message — the format NDMA's SACHET system ingests — and SMS text in English, Hindi and Gujarati."*
4:10–4:40 Validation	Validation — `/validation`	Scroll slowly from the headline stats to section 8	*"Every number is measured on storms and seasons the models never saw. We also publish what did not work: our track error is about twice IMD's operational forecasts, and satellite image sequences did not improve rapid-intensification warnings — so we kept the model that did."*

4:40–5:00 Close	Operations overview or Diagram 01	Hold on the full console	"*From a raw satellite image to a district alert — identification, classification and prediction in one decision-support tool, built on open data and designed to take IMD's INSAT imagery next. Thank you.*"

Recording tips: 1600×900 window, 100% zoom, light theme; OBS Studio or Xbox Game Bar (Win+G); record each segment separately and join; add captions for the numbers; keep the cursor slow; say "decision support", never "replaces IMD".

9 · PPT content, slide by slide (SIH template)

Your template has six slides. Replace the placeholder text as below. Diagrams are 16:9 PNGs: on a 16:9 slide, insert under the title and set width ≈ 12.3" (31 cm); on a 4:3 slide, width ≈ 9.3".

Slide 1 — Title page

- Problem Statement ID – **26070**
- Problem Statement Title – **AI/ML based system for identification, classification and prediction of different tropical cyclone patterns using multi-source satellite data**
- Theme – **Disaster Management** · PS Category – **Software**
- Team ID – (fill) · Team Name – NerveAi

Slide 2 — Idea title: "CycloneSense — from satellite pixel to district alert" (suggested) · Proposed solution

- **Identify early:** AI scans the whole Bay of Bengal and Arabian Sea satellite image every 3 h and finds cyclonic systems 24–42 h before they become cyclones.
- **Classify with reasons:** CNN estimates wind, Dvorak T-number and IMD category from IR imagery, with Grad-CAM showing which clouds drove the call.
- **Warn & predict:** rapid-intensification and genesis probabilities with explanations; 72 h track and intensity with a calibrated uncertainty cone.
- **Act:** landfall window, people exposed, fishermen sea-area warnings and district alerts as CAP 1.2 XML and multilingual SMS.
- **Innovation & uniqueness:** basin-wide identification + imbalance-aware RI warning + calibrated cone + cross-basin transfer learning (+14.9%) + alert-ready output, all replayable against real storms.

Optional image: docs/diagrams/02_end_to_end_flow.png (small, right side) or a dashboard screenshot.

Slide 3 — Technical approach

- **Technologies:** Python · PyTorch · timm (EfficientNet-B0, ResNet-18) · LightGBM · SHAP · xarray; FastAPI · SQLite · Docker; Next.js (React) · TypeScript · Tailwind · Mapbox GL · Recharts; Google Colab T4 GPU for training; Hugging Face Spaces + Vercel for hosting.
- **Data:** GridSat-B1 IR/WV, TCIR IR/WV/microwave, INSAT-3D (MOSDAC), IBTrACS, ERA5, WorldPop.
- **Methodology:** storm-centred crops → IBTrACS matching → IMD/Dvorak labels → storm/year splits → six models → decision layer → API → dashboard.

Diagram: docs/diagrams/01_system_architecture.png (full width). If allowed a 2nd technical slide: 03_training_validation.png or a dashboard screenshot as "working prototype".

Slide 4 — Feasibility and viability

1. **Feasibility:** working prototype already built on free, open data (NOAA, Copernicus, WorldPop, MOSDAC); trains on a free Colab GPU in ≈1.5 h; runs on a laptop CPU; Docker image deploys to free cloud tiers.
2. **Challenges & risks:** track skill below numerical weather models; sensor/processing shift between training data and INSAT; missing satellite channels; rare extreme storms; false alarms.
3. **Mitigation:** position as decision support with calibrated cone and replay; calibration and retraining on native MOSDAC data; channel-dropout training with IR-only fallback; cross-basin pretraining; thresholds tuned on validation years and 15 h track confirmation.

Diagram: docs/diagrams/08_risks_mitigation.png

Slide 5 — Impact and benefits

1. **Potential impact:** RSMC forecasters get earlier identification and RI warnings; SDMA/NDRF get landfall windows and exposure for pre-positioning; district collectors get staged, colour-coded alerts; fisherfolk get sea-area warnings in their language.
2. **Benefits — Social:** more lead time for evacuation ("Mission Zero Casualty"). **Economic:** better-targeted evacuations and fishing bans reduce losses and alert fatigue. **Operational:** standard CAP output plugs into existing alerting pipelines. **Regional:** scalable to the RSMC New Delhi area (Bangladesh, Myanmar, Sri Lanka, Pakistan, Oman).

Diagram: docs/diagrams/07_users_decisions.png

Slide 6 — Research and references

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6. Hersbach, H. et al. (2020). The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.*
7. Tan & Le (2019) EfficientNet, *ICML*; Shi et al. (2015) ConvLSTM, *NeurIPS*; Zhou et al. (2019) Objects as Points (CenterNet); Ke et al. (2017) LightGBM, *NeurIPS*; Lundberg & Lee (2017) SHAP, *NeurIPS*; Selvaraju et al. (2017) Grad-CAM, *ICCV*.
8. OASIS (2010). Common Alerting Protocol v1.2; NDMA SACHET; IMD RSMC New Delhi cyclone classification; WorldPop (2020) population counts.

Tip: add one line per reference saying how you used it, e.g. "TCIR — training data for the intensity CNN".

10 · Tech stack and datasets

Tech stack

Layer	Technology	Why
ML / training	Python 3.13, PyTorch 2.14, timm (EfficientNet-B0, ResNet-18), LightGBM 4.7, scikit-learn, SHAP	Proven CV backbones; fast, explainable tabular models
Geo / data	xarray, netCDF4, h5py, pandas, pyarrow, rasterio, shapely, cdsapi, kaggle	Read satellite HDF5/NetCDF, reanalysis and GIS data
Training compute	Google Colab T4 GPU (≈1.5 h total); laptop CPU for tabular models	Free, reproducible notebooks in `notebooks/`
Backend	FastAPI, Uvicorn, SQLite audit log, optional Gemini API (guard-railed)	Async, auto-generated API docs at /docs
Frontend	Next.js 16 (React 19), TypeScript, Tailwind CSS 4, Mapbox GL JS 3, Recharts 3, lucide icons	Fast single-screen console, neumorphic UI, light/dark
Deployment	Docker, Hugging Face Spaces (backend), Vercel (frontend), GitHub	Free tiers; one-command local run

Datasets

Dataset	Source	Content	Used for
TCIR (CPAC + IO + SH)	Chen et al. 2018; Kaggle mirror	≈23,000 storm-centred frames 2003–16, 201×201 at 0.07°, IR / WV / VIS / microwave, 3-hourly, intensity labels	Intensity CNN, multi-channel model, ConvLSTM
GridSat-B1	NOAA NCEI	Global geostationary IR + WV, 3-hourly, 0.07°; ≈1,000 scenes subset to the NIO box	Basin detector, demo scenes
INSAT-3D cyclone images	Kaggle (sshubam), MOSDAC-sourced	136 labelled IR images, 2013–21	External test, upload demo
IBTrACS v04r01 (NI)	NOAA NCEI	450 storms 1982–2025: track, JTWC and IMD winds, pressure	Labels, RI, genesis, track, backtests
ERA5 monthly means	Copernicus CDS	SST, 200/850 hPa winds (shear), 600 hPa humidity, 1982–2025	RI and genesis environment
WorldPop 2020 (1 km)	WorldPop	Population of India, Bangladesh, Myanmar, Sri Lanka, Pakistan	Exposure
Natural Earth 10 m land	Natural Earth	Coastline / land mask	Landfall detection

Training & validation — built so judges can trust the numbers

Every model is scored on storms or seasons it never saw; splits are by storm and year, never random frames

Model	Training data	Tuned on	Tested on (never seen)	Where trained
Intensity CNN + Dvorak head	TCIR: Pacific & S. Hemisphere (pretrain) → NIO ≤2014	NIO 2015	**NIO 2016** (Vardah, Kyant, Nada, Roanu) + 132 INSAT-3D images	Colab T4 GPU
Multi-channel CNN IR/WV/PMW	same, with random channel dropout	NIO 2015	**NIO 2016**	Colab T4 GPU
Basin detector	339 GridSat scenes 2005–15 (+ storm-free scenes)	2016	**189 scenes 2017–24** + 6 demo storms	Colab T4 GPU
ConvLSTM sequences	TCIR 8-frame sequences, all basins	held-out storms	**NIO 2016** + all-basin validation	Colab T4 GPU
RI early warning	IBTrACS NIO 1982–2016 (15:1 imbalance)	2017–19	**2020–25** (Amphan, Tauktae, Mocha, Biparjoy…)	Laptop CPU
Genesis	IBTrACS depressions 1982–2016 + ERA5	2017–19	**2020–25**	Laptop CPU
Track + cone	IBTrACS NIO 1982–2016	2017–19 (cone calibrated here)	**2020–25**	Laptop CPU

Class imbalance handled explicitly: RI is 1 in 15 fixes → class-weighted training + probability calibration; threshold chosen on validation years only.

Cross-basin transfer for data-scarce NIO: pretraining on other ocean basins cut intensity error by **14.9%**.

No look-ahead: environment uses the previous month's ERA5; every replayed forecast uses data up to that time only.

Figure 7 — Training and validation splits (no leakage)

11 · Going live: hosting the prototype

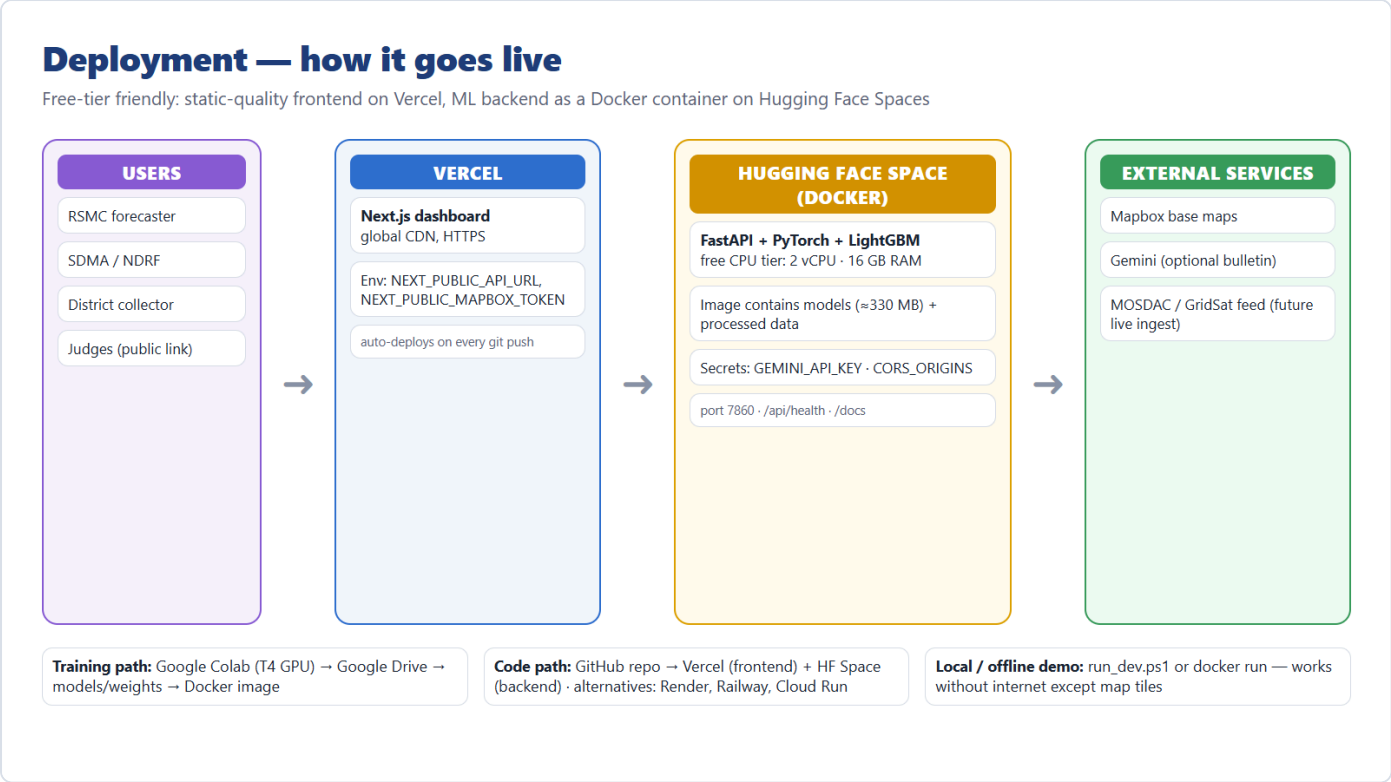


Figure 8 — Deployment architecture

First — security. Several keys were shared in chat during development. Before making anything public, regenerate: Kaggle token, Gemini API key, Copernicus CDS key, Mapbox token, and change the NASA Earthdata password. Never commit `.env` (it is already in `.gitignore` and `.dockerignore`).

A · Backend on Hugging Face Spaces (free: 2 vCPU, 16 GB RAM)

1. Create an account at huggingface.co → **New Space** → SDK: **Docker** → template: Blank → hardware: **CPU basic (free)** → visibility Public or Private.
2. Install Git LFS (git-lfs.com), then clone the Space and copy the backend files into it:

```
git clone https://huggingface.co/spaces/<user>/cyclone-api
cd cyclone-api
git lfs install
git lfs track "*.pt" "*.npz" "*.parquet" "*.nc" "*.npy" "*.db"
# copy from the project folder: Dockerfile, .dockerignore, ml/, backend/,
# models/weights/ (skip *_last.pt and MC_*_best.pt), data/processed/, data/raw/era5/
git add . && git commit -m "Cyclone API" && git push
```

3. In the Space → **Settings** → **Variables and secrets**: add secret `GEMINI_API_KEY` (optional) and variable `CORS_ORIGINS` = your Vercel URL, e.g. `https://cyclone-dss.vercel.app`.
4. Wait for the build (≈10–15 min the first time). Test `https://<user>-cyclone-api.hf.space/api/health` and `/docs`.
5. Free Spaces sleep after inactivity: open the URL a few minutes before a demo.

B · Frontend on Vercel (free)

1. Push the project to a GitHub repository (the `frontend/` folder is enough; model weights are not needed by the frontend).

2. vercel.com → **Add New** → **Project** → import the repo → **Root Directory:** `frontend` (framework auto-detected as Next.js).
3. Environment variables: `NEXT_PUBLIC_API_URL` = the Space URL (no trailing slash); `NEXT_PUBLIC_MAPBOX_TOKEN` = your public Mapbox token.
4. Deploy. In Mapbox account → Tokens, restrict the token's URLs to your Vercel domain.
5. Every `git push` redeploys automatically.

C · Test the image locally first (optional)

```
docker build -t cyclone-api .
docker run -p 7860:7860 -e CORS_ORIGINS=http://localhost:3000 -e GEMINI_API_KEY=... cyclone-api
# then set frontend/.env.local NEXT_PUBLIC_API_URL=http://localhost:7860
```

D · Alternatives

Option	Cost	Notes
Hugging Face Spaces (recommended)	Free	16 GB RAM fits PyTorch + models; sleeps when idle
Google Cloud Run	Free tier, then pay per use	Scale to zero; set memory 2–4 GB; deploy the same Dockerfile
Railway / Render	≈$5–7 / month	Need ≥2 GB RAM; free 512 MB tiers are too small for PyTorch
Own laptop + hotspot	Free	`run_dev.ps1`; dashboard reachable on the LAN IP; best fallback for the finale

E · Post-deployment checklist

- /api/health shows 4 core models; /api/basin/storms shows 6 demo storms.
- Dashboard loads all five pages; map tiles appear (Mapbox token + URL restriction correct).
- No CORS errors in the browser console (CORS_ORIGINS matches the Vercel URL exactly).
- Warm the basin scans once (open each demo storm) so the judges see instant results.
- Keep the local setup and a recorded video as backups.

12 · Judge Q&A preparation

Likely question	Suggested answer
"What is your accuracy?"	"We measure each task the way forecasters do, on storms the models never saw: wind error about 7 knots, rapid-intensification warning caught all 8 RI storms of 2020–25, storms identified 24–42 hours before cyclone stage, and a cone that holds 66% of real positions against a 67% target."
"Is your track better than IMD's?"	"No — about twice IMD's operational error, because we use track history only, without numerical weather model steering winds. It is decision support; adding GFS steering flow is our next step."
"Did you use INSAT data?"	"Yes, 132 labelled INSAT-3D images as an external test. Error rises to about 15 kt after calibration because of different image processing — so production training should use native MOSDAC HDF5, which our pipeline is ready for."
"How do you handle class imbalance?"	"RI is 1 in 15 observations. We use class weighting, calibrate probabilities, choose the threshold on 2017–19 only, and report PR-AUC and false alarms, not accuracy."
"How do you prevent data leakage?"	"All splits are by storm and year. Tabular models: train to 2016, tune 2017–19, test 2020–25. Image models test on 2016+ storms; the detector on 2017–24 scenes."
"Is it connected to SACHET?"	"No. It produces CAP 1.2, the standard SACHET ingests, marked status = Exercise. A pilot with NDMA/C-DOT would be the next step."
"What if a satellite channel is missing?"	"Models are trained with channel dropout. GridSat 2024 has no water vapour: the system switched to IR-only intensity and still detected Cyclone Dana in 80% of fixes."
"Why did the image-sequence model not help?"	"We tested it fairly against the same network without images: better 24-hour change error, worse RI ranking. So the operational RI warning uses the model that performs best, and we report the negative result."
"Can it run in real time?"	"Inference takes about one to three seconds per forecast or scene on a laptop CPU. Real-time operation needs a live INSAT or EUMETSAT feed instead of the GridSat archive; the ingest slot is designed for MOSDAC."
"Why use an LLM?"	"Only to word an optional bulletin from the model's numbers. It is constrained, falls back to a template, and strips any official-agency headers."

Limitations (state them proactively)

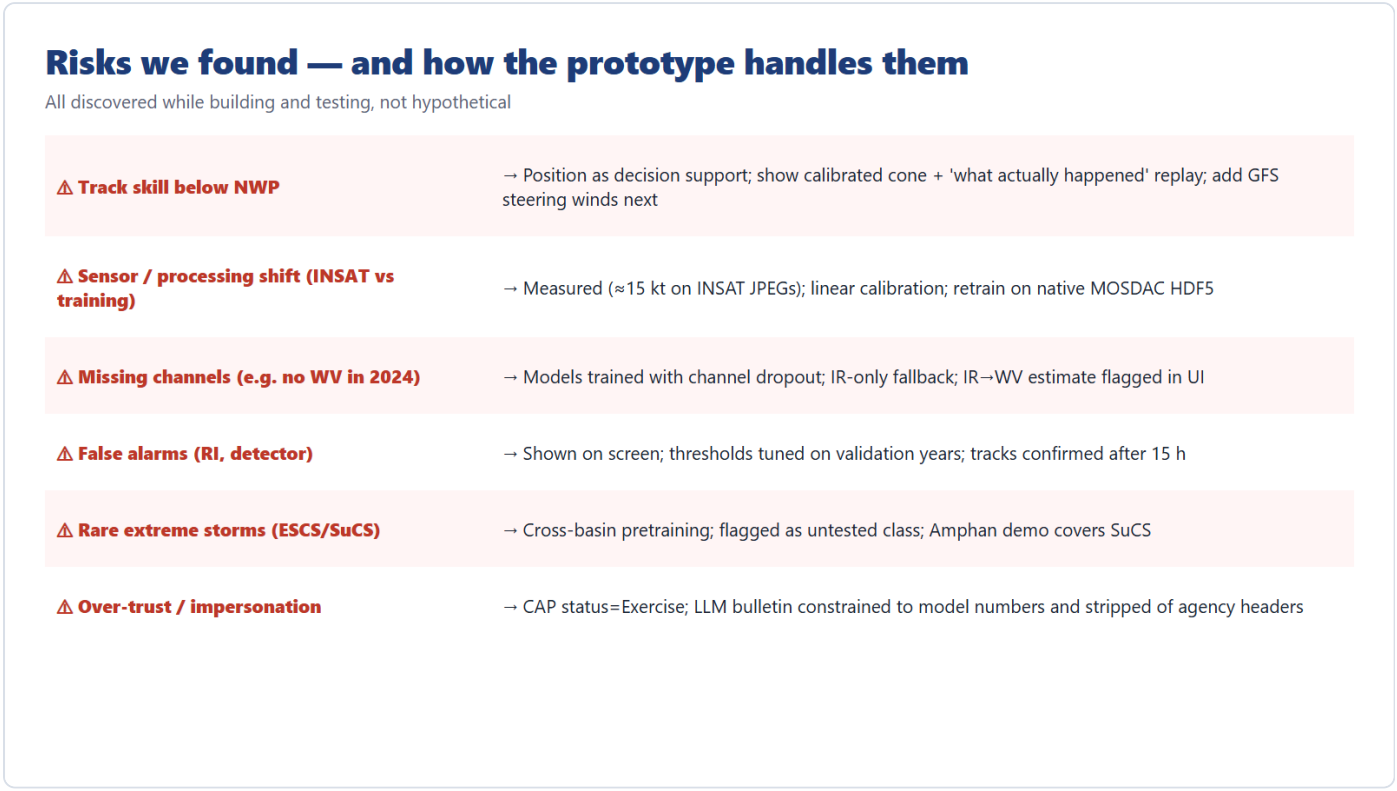
- Decision support only — not an official IMD/NDMA product; CAP status is Exercise.
- Track skill ≈2× IMD operational error; skill falls quickly beyond 48 h.
- Domain shift on INSAT JPEGs; no ESCS/SuCS frames in the satellite test year.
- RI warning also flags some non-RI storms (7 of 19); detector recall is lower for weak depressions.
- District points and gale radii are illustrative approximations.

Future scope

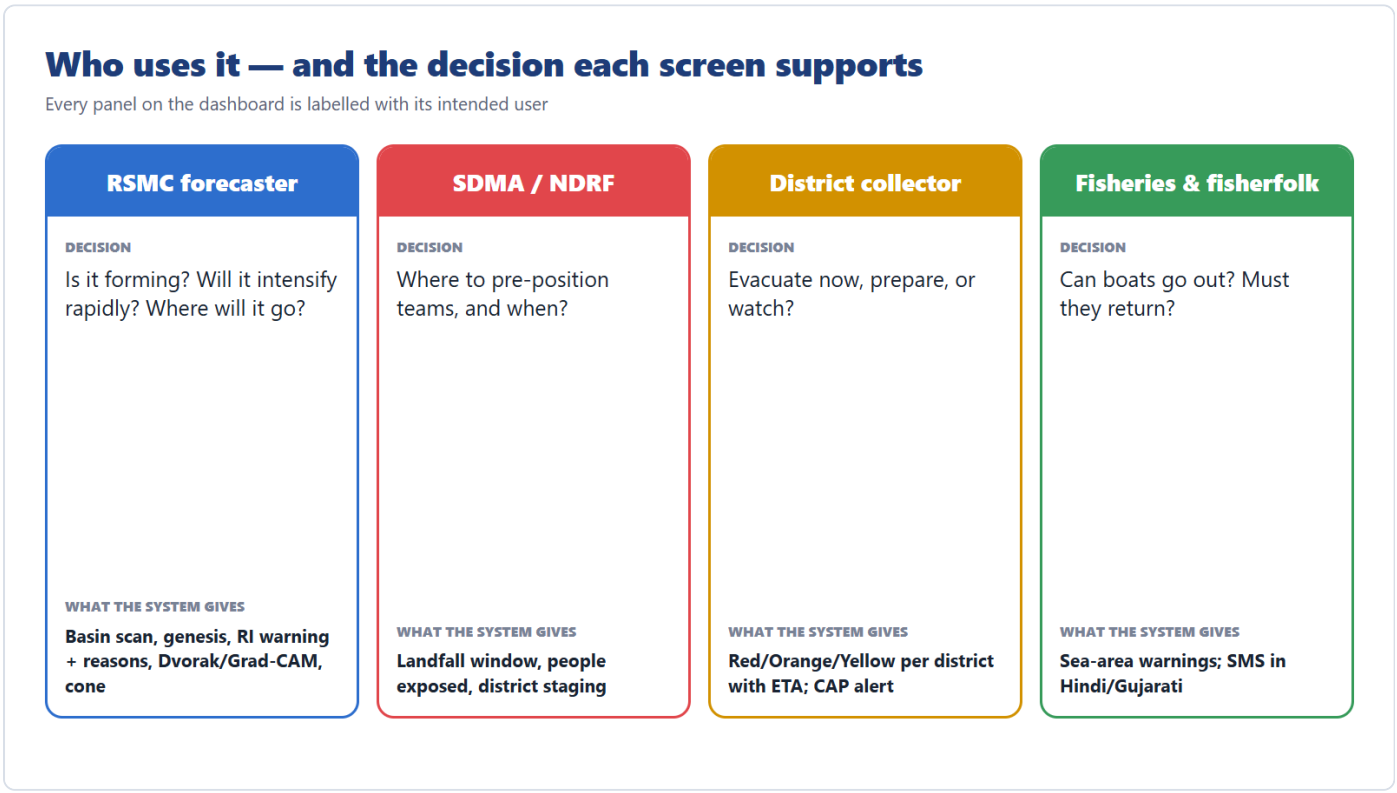
- Live INSAT-3D/3DR/3DS ingest from MOSDAC and retraining on native HDF5.
- GFS/ERA5 steering winds and daily environment fields for track and RI.
- CAP pilot with NDMA/C-DOT SACHET; more languages (Odia, Tamil, Telugu, Bengali, Marathi).
- Extension to the whole RSMC New Delhi area; foundation-model track forecasting; physics-informed augmentation for rare extreme storms.

Appendix · Diagrams for the PPT

Each file in `docs/diagrams/` is 1600×900 px (16:9). Suggested placement: 01 → Slide 3 · 02 → Slide 2 · 03 → Slide 3 (2nd) · 04 → results/Slide 4 · 05 → Slide 2 or 5 · 06 → deployment/Slide 4 · 07 → Slide 5 · 08 → Slide 4 · 09 → video planning.



08_risks_mitigation.png



07_users_decisions.png